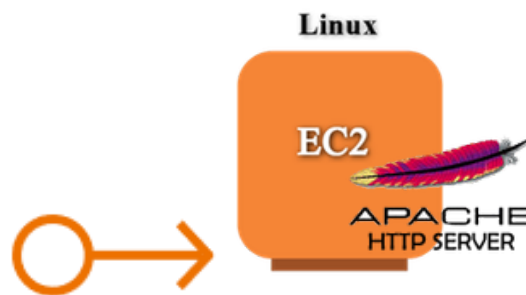




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator

6.Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

1. Highly Available Applications:

- o If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.

2. Web Servers:

- o If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.

3. Disaster Recovery:

- o Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

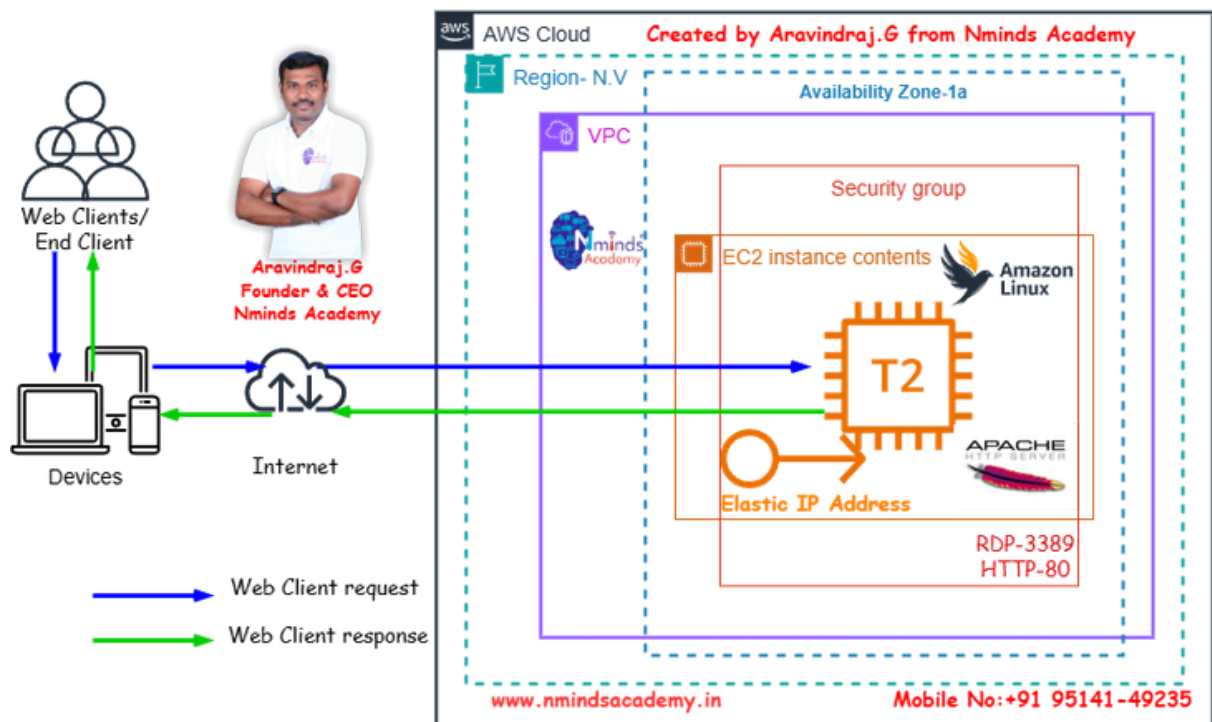
Best Practices:

- o Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- o Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- o Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.



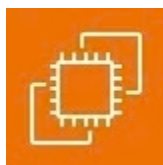
Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS
Services in Windows Server

Used



Amazon EC2



Apache Webserver



Amazon ElasticIPAddress



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Execution Tasks:

The screenshot shows the 'Allocate Elastic IP address' page in the AWS console. The breadcrumb navigation is 'EC2 > Elastic IP addresses > Allocate Elastic IP address'. The page title is 'Allocate Elastic IP address info'. Under 'Elastic IP address settings', the 'Public IPv4 address pool' is selected, with the option 'Amazon's pool of IPv4 addresses' chosen. The 'Network border group' is set to 'us-east-1'. A section for 'Global static IP addresses' mentions AWS Global Accelerator. A 'Tags - optional' section is also present. The footer includes 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services, Inc.

The screenshot shows the 'Elastic IP addresses' page in the AWS console. A green banner at the top states 'Elastic IP address allocated successfully. Elastic IP address 44.210.222.117'. The page title is 'Elastic IP addresses (1) info'. A table lists the allocated IP address. The left sidebar shows the navigation menu with categories like 'Images', 'Elastic Block Store', and 'Network & Security'. The 'Network & Security' section is expanded, showing 'Elastic IPs' as the selected option.

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated
-	44.210.222.117	Public IP	eipalloc-07d3d2cd36d2551d9	-	-



Execution Tasks:

The screenshot shows the AWS Management Console interface. At the top, a green notification banner states: "Elastic IP address associated successfully. Elastic IP address 44.210.222.117 has been associated with instance i-0813ccc1fc44a9dd0". Below this, the "Elastic IP addresses (1)" section is visible. It includes a search bar and a table with one entry:

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated Instance
-	44.210.222.117	Public IP	eipalloc-07d3d2cd36d2551d9	-	i-0813ccc1fc44a9dd0

Below the table, there is a section "Select an elastic IP address" with a link to "View IP address usage and recommendations to release unused IPs with Public IP insights". The left sidebar shows the navigation menu with categories like Images, Elastic Block Store, and Network & Security. The bottom of the console shows the footer with "© 2026, Amazon Web Services, Inc. or its affiliates."

The screenshot shows a web browser displaying a personal website. The page has a teal header and footer. The main content area is white and contains the following sections:

About Me

Hello!

My name is **Shabhari Vaasun**. I am a Computer Science student interested in programming, machine learning, and software development. I enjoy building projects that solve real-world problems and improve people's lives.

Education

Currently pursuing B.E in Computer Science and Engineering (AI & Data Science). I am learning programming languages like C, C++, Java, and Python along with web technologies.

Skills

C C++ Java Python HTML CSS Machine Learning

Projects

- Weather Prediction Application
- Smart Home Energy Usage Tracker
- Campus Placement Prediction using Machine Learning

Contact

Email: yourmail@example.com
Location: Tamil Nadu, India

